The Architecture of Intent: Pre-Motor Kinematic Telemetry for Real-Time Human-Robot Synchronization

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Technical White Paper | Specification v1.0
Module 01: Robotics Edition

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Client Repository: github.com/Neuropatic-Labs/si-module-01-bridge

I: The Actuator Inertia Bottleneck

Every modern commercial service robot, autonomous handoff station, and interactive kiosk deployed today shares a fundamental flaw: it is biologically blind. Over the past decade, robotics has experienced a golden age of digital reasoning. Large multimodal models can parse visual scenes, identify individual objects, and plan abstract task sequences with remarkable semantic fidelity. Yet the moment a service manipulator or interactive station interfaces directly with a person at a shared counter or handoff zone, this intelligence dissolves into awkward mechanical hesitation. The reason is simple: our machines are trapped in an entirely reactive control loop. When a person reaches out to receive an order, pass an item, or interact with a station, the machine waits. It waits for the physical hand to cross a predefined spatial perimeter. It registers the breach, computes a fresh trajectory, clears its collision checks, and only then commands its actuators to spool up and move.

To the human user, this sequence feels jarring and unnatural. The machine pauses, stutters, and then jerks forward. This hesitation is not just an aesthetic flaw; it is a primary operational bottleneck that caps cycle throughput, degrades customer experience, and disrupts fluid human-machine cadence.

Reactive Paradigm (Current Failure Mode):

[Human Hand Enters Zone] ──► [Sensor Triggers] ──► [Trajectory Planner] ──► [Motor Spin-Up] ──► [Robot Motion]
▲──────────────────▲
150–300 ms Lag

The Unforgiving Reality of Physical Mass

Robotics engineers routinely attempt to solve this interaction stutter by upgrading sensor frame rates or optimizing compute loops from 10 milliseconds down to 2 milliseconds. But shaving micro-cycles off path-planning algorithms fails to address the dominant physical bottleneck: **mechanical inertia.** Silicon computes in nanoseconds; electric actuators operate under the laws of classical physics. A standard 6-DoF commercial robotic joint does not reach operating velocity instantaneously. Between the issuance of a current command and the actual overcoming of rotor inertia, gear friction, and payload mass, there exists a rigid mechanical barrier:

$$\Delta t_{inertia} \approx 150 \text{ to } 300 \text{ ms}$$

During this 150–300 ms window, the robot arm remains effectively stationary while internal motor coils build torque. Because the system waited for the human hand to arrive before commanding motion, that mechanical spin-up time is stolen directly from the interaction cadence. The human is already waiting at the exchange point while the machine is only beginning its physical acceleration. You cannot code your way out of mechanical inertia by tuning motor gains. Over-accelerating to compensate creates dangerous torque spikes, end-effector vibration, and safety halts. If a machine is to interact with humans with natural, fluid cadence, it cannot wait for the physical hand to arrive. It must possess an anticipatory instinct. It must begin spooling its motors before the human limb ever leaves its resting trajectory. Anticipatory torque ramp-up does not command sudden full-speed displacement; rather, it overcomes static friction and initiates smooth, low-velocity pre-alignment trajectories well within standard commercial service velocity profiles.

II: Evolutionary Predictive Mechanics

Nature solved the latency problem of multi-agent physical coordination millions of years ago. In biological nervous systems, pure reflex-based control is too slow to support complex manipulation or close-quarters collaboration. Transmitting sensory data from the retina or periphery, routing it through cortical processing, and propagating efferent motor spikes down to muscle fibers incurs a biological transmission penalty of 100 to 180 ms. If humans coordinated reactively, two people handing off a simple tool would stutter, hesitate, and drop objects just as modern robotic cells do.

Instead, human motor control relies fundamentally on an internal predictive forward model. We do not wait for another person's hand to arrive at an exchange coordinate; we read early physiological signatures that forecast spatial intent well before the ballistic limb completes its reach.

Ocular Pre-Fetch: The Land & Hayhoe Paradigm

The primary sensory pre-fetch mechanism in human physical interaction is the temporal decoupling between ocular fixation and limb kinematics. Classic motor-control and visual perception literature (notably Land & Hayhoe, as well as foundational studies in human spatial attention) demonstrates that during targeted object manipulation or cooperative handoffs, human gaze does not track the hand. Rather, gaze acts as an advance spatial scout. In goal-directed reaching tasks, the visual system executes a saccade and secures a stable foveal lock on the target coordinates hundreds of milliseconds before the arm begins accelerating along its trajectory. To prevent ambient visual scanning from triggering false-positive actuation, Module 01 enforces a dynamic spatio-temporal foveal dispersion filter: intent directives are only emitted when ocular fixation stability satisfies strict solid-angle dispersion and pre-motor dwell thresholds on an active interaction target.

$$\Delta t_{advance} = t_{reach_onset} - t_{fixation_lock} \approx 200 \text{ to } 300 \text{ ms}$$

During this advance window, the biological visual system extracts target geometry, spatial clearance, and orientation vectors, transmitting these parameters to motor planning centers in the premotor cortex. By the time physical kinematics manifest at the wrist and fingertips, the spatial destination is already established and immutable.

Optical Degrade & Graceful Fallback:
When direct ocular vectors encounter edge-case optical occlusion (e.g., eyewear reflections or acute head angles), Module 01 seamlessly weights ballistic wrist/hand velocity vectors to maintain pre-motor trajectory estimation without pipeline stall.

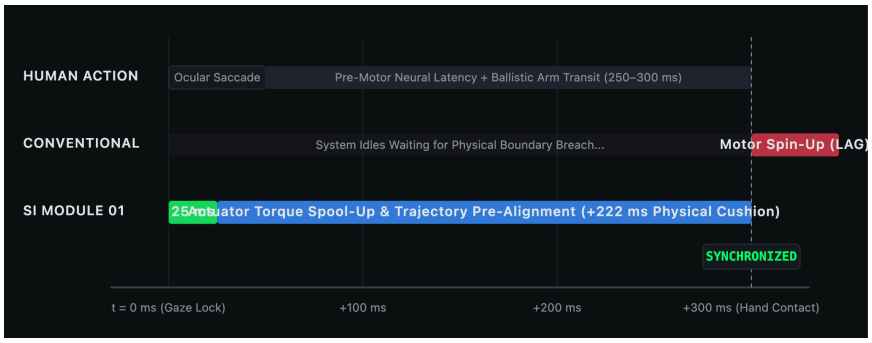


Figure 1. Chronological pre-motor latency budget: Ocular gaze pre-fetch delivers a +222 ms physical head start to actuators prior to ballistic limb arrival.

Bringing Biological Cadence to Synthetic Controllers

The fundamental failure of conventional physical automation is that machines throw away this advance biological signaling window. They treat human motion as an arbitrary point-cloud intrusion rather than a structured, highly predictable physiological chain of events. By the time a commercial manipulator's depth camera registers a hand breaching an interaction envelope, the human's visual system has already been locked on the exchange zone for nearly half a second.

Traditional Sensor Pipeline vs. Symbiotic Intelligence Layer:

Conventional: [Human Gaze Lock] ──► [Limb Enters Zone] ──► [Machine Detects & Spools Actuators] (Lag)

SI Layer: [Human Gaze Lock] ──► [Machine Spools Actuators] ──► [Synchronized Handoff] (Fluid)

The Symbiotic Intelligence Layer leverages this evolutionary pre-motor mechanic. Treating the human user's ocular fixation as a real-time spatial pre-fetch signal and coupling it with early kinematic transit vectors, the software synthesizes a continuous, deterministic intent directive. This gives the downstream robotic controller a 200 to 300 ms temporal cushion. Within this advance window, the machine can:

- Overcome actuator inertia and spool up joint motors before the hand enters the exchange zone.
- Pre-align end-effector orientation to match the approaching hand's trajectory.
- Eliminate the hesitation stutter entirely, transforming awkward stop-and-go machinery into an intuitive, synchronized partner.

Crucially, implementing this capability does not require redesigning existing robotic hardware, replacing motor drives, or running invasive wearable sensors. It requires only an edge-native telemetry pipeline capable of decoding these natural non-contact optical signals in real time.

III: Time-Budget Analysis & Edge Architecture

To bridge pre-motor physiological signaling with physical robot kinematics, the execution pipeline must operate within a deterministic time budget. Every millisecond consumed by image capture, inference, and serialization directly erodes the advance buffer required by the robot's physical motors. Module 01 solves this by executing entirely as an on-premise, compiled edge runtime that ingests non-contact optical telemetry and outputs discrete intent directives over a standard 60Hz WebSocket interface.

The End-to-End Latency Budget

The following table breaks down the chronological sequence from the instant an operator fixates on an interaction target to the synchronized physical arrival of the end-effector:

Event Phase	Elapsed Time	Action & Pipeline State	Advance Buffer Remaining
t_0	**0.0 ms**	**Ocular Fixation Criterion Met:** User's foveal gaze satisfies spatio-temporal dwell threshold on target exchange coordinate.	+250.0 ms (Biological Lead)
t_1	**16.6 ms**	**Worst-Case Sensor Frame Ingestion:** Edge optical sensor captures next 60Hz frame boundary.	+233.4 ms
t_2	**24.1 ms**	**Vector Extraction & Temporal Coupling:** Module 01 resolves 3D gaze ray and checks hand reach onset (7.5 ms compute).	+225.9 ms
t_3	**25.3 ms**	**Directive Serialization:** System packages JSON directive and transmits via local loopback WebSocket (1.2 ms socket transit).	+224.7 ms
t_4	**28.0 ms**	**ROS 2 Path Update:** Client node (`si_module_01_bridge.py`) receives packet; motion planner updates trajectory goal (2.7 ms).	+222.0 ms (NET HEAD START)
t_5	**35.0 ms**	**Actuator Torque Command:** Inverter stages energize; joint motors begin overcoming physical inertia and gear friction.	+215.0ms
t_6	**210.0 ms**	**Actuators Reach Operating Velocity:** Joint motors overcome mechanical inertia (approx 175 ms spool-up window).	+40.0 ms
t_7	**250.0 ms**	**Human Ballistic Limb Arrival:** Human hand enters the exchange envelope. End-effector is already pre-aligned and decelerating into position.	**0.0 ms (Synchronous Cadence)**

Delivering a net physical head start exceeding +220 ms, the system fully absorbs mechanical motor inertia (Delta t_inertia approx 150–300 ms) within the human pre-motor window. The machine accelerates concurrently with the operator's intention rather than reacting to the physical handoff.

The Clean Edge Boundary

Module 01 acts strictly as an anticipatory telemetry provider, maintaining a clean architectural decoupling from client motion controllers. Module 01 is strictly an advisory intent provider. It does not bypass client-side collision avoidance, hardware E-stops, or functional safety kernels. Downstream planners retain final kinematic arbitration:

- **Zero Hardware Intrusion:** Operates via non-contact optical sensing. No specialized wearables, force-torque gloves, or proprietary sensor rigs are required.
- **Controller Agnostic:** Downstream systems receive standard JSON directives over local WebSockets. Whether the robot runs MoveIt 2, custom C++ velocity controllers, or proprietary vendor trajectory planners, the integration contract remains identical.
- **Immutable Edge Binary & Irreversible Extraction:** All spatial projection heuristics, neural gaze-cone estimators, and temporal filters are compiled into optimized machine-code libraries (.so) running inside an isolated Docker runtime. The internal mathematical weights and calibration matrices cannot be reconstructed from external calls, and raw camera inputs are transformed into non-biometric unit vectors before reaching memory.

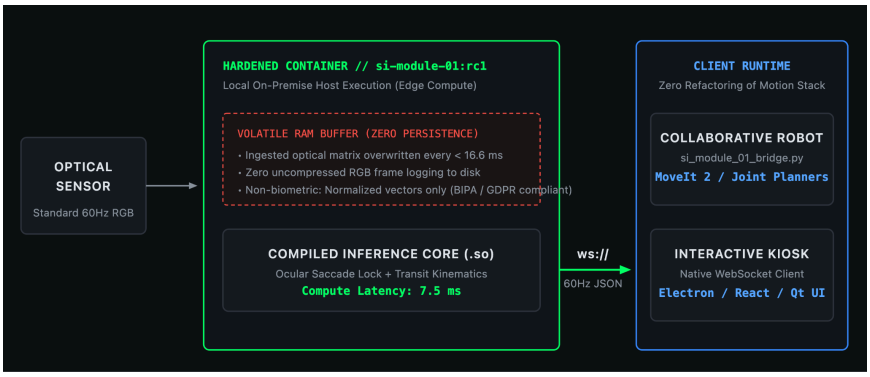


Figure 2. Edge telemetry architecture: Volatile RAM ingestion buffer, sub-17 ms frame flush boundary, and isolated 60Hz JSON client distribution over local loopback.

Decoupled Station Sensing Topology: Ambient Infrastructure vs. Eye-in-Hand

A frequent failure mode in service manipulation is the reliance on robot-mounted perception (e.g., wrist-mounted "eye-in-hand" cameras or mobile head cams). When a robotic manipulator turns away from a person, such as rotating to retrieve an item, service a rear preparation counter, or clear a staging tray, robot-mounted sensors lose visual line of sight with the interaction zone. Re-establishing perception only after the manipulator returns to the handoff boundary forces the motion planner into an immediate cold-start, re-introducing reactive actuator hesitation. Module 01 fundamentally decouples human intent perception from robot kinematic state by operating over ambient station infrastructure sensing:

- **Unbroken Visual Line of Sight:** The edge runtime ingests frames from a static, external workspace optical sensor (such as an off-the-shelf wide-angle USB/RTSP camera) positioned with persistent visual coverage over the shared collaboration zone.
- **Continuous Asynchronous State Updates:** Even while downstream actuators are executing non-collaborative motions with their backs turned, Module 01 continuously tracks gaze vectors, dwell transitions, and pre-motor reaching arcs in the background at 60 Hz.
- **Zero-Re-Tooling Hardware Overhead:** Fleet operators and robotics OEMs do not need to modify internal robot harnesses, redesign end-effectors, or alter primary compute payloads. Intent telemetry is streamed entirely out-of-band across the local edge network, allowing motion planners to ingest pre-oriented target directives the moment the robot begins its return trajectory.

The Volatile Memory Guarantee (Privacy & Compliance)

Deploying optical vision systems into commercial shared environments, whether autonomous service counters, hospitality handoff stations, or interactive kiosks, demands strict compliance with biometric liability frameworks (including BIPA and GDPR). Module 01 is architected under an immutable non-retention protocol:

1. **Volatile RAM Pipeline:** Ingested image matrices reside solely in volatile RAM buffer registers for the duration of vector extraction.
2. **Sub-17 ms Frame Flush:** As each new 60Hz optical frame arrives, the preceding raw buffer is overwritten and purged. The system maintains no circular video logs, uncompressed frame archives, or local disk caches.
3. **Ephemeral Vector Abstraction (Zero Biometric Identity):** The inference engine operates exclusively as a geometric vector extractor. It does not compute, store, or output biometric identifiers, facial geometry templates, iris scans, or demographic markers. Once normalized 3D unit gaze vectors and hand velocity vectors are computed, all underlying pixel frames in volatile memory registers are completely purged within the sub-17 ms execution window. The software leaves zero persistent telemetry footprint on host hardware.
4. **Zero Cloud Egress:** All compute executes locally on the host edge hardware. No image frames, telemetry metadata, or diagnostic pings leave the local machine.

This guarantees that operators deploy the anticipatory speed of vision-guided control without inheriting data privacy risks or customer surveillance liabilities.

IV. Telemetry Contract & ROS 2 Integration

To ensure rapid integration into commercial fleets, Module 01 enforces a strict, decoupled boundary between perception telemetry and downstream motion control. The engine does not usurp low-level joint velocity controllers or inverse kinematics solvers; instead, it outputs structured pre-motor directives that existing planners can consume natively.

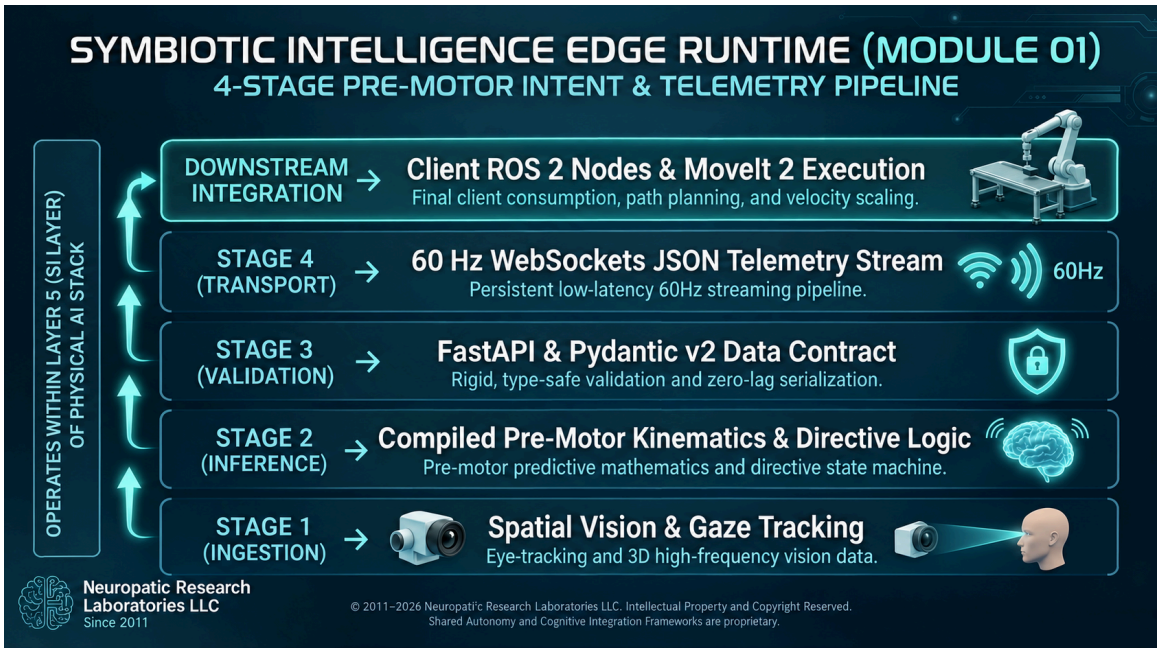


Figure 3. Symbiotic Intelligence Edge Runtime Architecture (Module 01)
End-to-end edge pipeline illustrating multi-modal pre-motor intent ingestion (60 Hz), strict Pydantic v2 data serialization, low-latency WebSocket streaming (< 2 ms transit), and downstream ROS 2 trajectory planning integration.

The 60Hz Telemetry Schema

The edge container exposes a local WebSocket server at `ws://127.0.0.1:8000/ws`. At every 60Hz tick, the client runtime ingests a serialized JSON telemetry packet:

```
{
  "timestamp": 1727280000.1842,
  "frame_id": 4021,
  "gaze_vector": [0.142, -0.051, 0.988],
  "target_xy_normalized": [0.684, 0.412],
  "target_identifier": "HANDOFF_TARGET_A",
  "gaze_locked": true,
  "reach_transit_active": true,
  "intent_directive": "PREEMPTIVE_ORIENT_AND_EXTEND",
  "measured_lead_time_ms": 234.0,
  "confidence": 0.95
}
```

Field Definitions:

The 60Hz payload distributes normalized 3D unit gaze vectors (gaze_vector), calibrated planar intersection coordinates (target_xy_normalized), and discrete interaction zones (target_identifier). The engine flags gaze_locked upon spatio-temporal foveal dwell verification, and activates reach_transit_active upon early ballistic reach detection. Downstream motion controllers ingest intent_directive (`IDLE, PREEMPTIVE_ORIENT_AND_EXTEND, HANDOFF_ZONE_OCCUPIED, ACTIVE_DECELERATION, HOLD`) alongside the clocked pre-motor lead-time buffer (measured_lead_time_ms) and spatial confidence (0.00 to 1.00).

Downstream ROS 2 Architecture

For robotic manipulators and mobile bases operating on ROS 2 (Humble, Iron, or Jazzy), the translation layer is handled by the open-source client node: `si_module_01_bridge.py`.

Client Adapter Implementation

The bridge node maintains an asynchronous WebSocket client connection to the local container, validates incoming 60Hz telemetry packets against the wire schema and watchdog timer, and re-broadcasts deterministic directives onto /si_layer/module_01/intent_directive for consumption by trajectory planners. The reference Python ROS 2 bridge node (si_module_01_bridge.py) and synthetic 60Hz telemetry replay server (mock_telemetry_server.py) are available in the public integration repository at: https://github.com/Neuropatic-Labs/si-module-01-bridge (with production C++ bindings provided in the enterprise evaluation pilot stack).

Trajectory Planner Integration (MoveIt 2)

By subscribing to `/si_layer/module_01/intent_directive`, motion planning stacks bypass passive perimeter waiting loops:

1. **State Trigger:** Upon receiving `PREEMPTIVE_ORIENT_AND_EXTEND`, the planner reads `target_identifier` and begins solving inverse kinematics for the pre-approach pose.
2. **Inertia Compensation:** The robot commands low-velocity joint acceleration toward the exchange zone while the human's hand is still in transit.
3. **Safety Guarantee:** If `confidence` drops or an anomalous trajectory triggers `ACTIVE_DECELERATION`, the motion planner instantly transitions into compliant impedance float or freezes trajectory execution within standard safety limits.
4. **Watchdog Heartbeat Guarantee:** The container emits a continuous monotonic frame_id at 60Hz. If the client adapter detects a packet dropout exceeding 33.3 ms (2 consecutive frames), the planner immediately triggers a graceful, velocity-damped trajectory hold to guarantee workspace safety.

This integration model allows robotics engineering teams to eliminate interaction lag in their deployments within an afternoon, without refactoring core motion planners or altering their underlying safety architecture.

V: The Category Horizon: Anticipatory Physical Automation

The ultimate destiny of human-machine interaction is not faster reactive algorithms; it is symbiotic anticipation. For decades, physical automation has remained passive until physical boundary breaches or contact force a response. By equipping synthetic controllers with evolutionary pre-motor foresight, Module 01 eliminates the friction dividing human intent from mechanical execution.

Biological nervous systems do not wait for physical impact to decode direction; they operate on an evolutionary forward model that telegraphs spatial intent hundreds of milliseconds before contact occurs. Module 01 delivers the first commercial primitive for true physical foresight: an on-premise, zero-liability edge runtime that bridges human pre-motor intent and synthetic physical execution.

Whether embodied as a 6-axis manipulator pre-aligning for a seamless service handoff or an autonomous mobile manipulator clearing a shared workspace cell, machines will no longer wait for physical contact to begin moving. They will meet human cadence in real time.

Evaluation Pilot Stack & Enterprise Licensing

Neuropatic Research Laboratories LLC provides qualified robotics manufacturers and automation integrators a structured **90-day on-premise evaluation pilot**. The pilot deployment includes:

- **Hardware Container Distribution:** Pre-compiled, cryptographically signed si-module-01:rc1 edge container configured for your local compute environment.
- **Client Integration Stack:** Open-source ROS 2 Python bridge adapter (si_module_01_bridge.py), production C++ bindings, and MoveIt 2 trajectory configuration templates.
- **On-Site Telemetry Calibration:** Zero-intrusion optical baseline calibration against existing actuator cycle times to verify lead-time gains in your physical deployment environment.

Initiate Evaluation Sandbox:
Contact Systems Architecture: pilots@neuropatic.com
Documentation & Client Repository: https://github.com/Neuropatic-Labs/si-module-01-bridge